SK_Scout:

A Scalable Framework for Detecting Coordinated Behavior in GitHub Activity Streams

Github Link: https://github.com/shravanips/SK_Scout

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AIM AND MOTIVATION

GitHub is a platform of interest for developers and learners across the globe. With the growing use of Artificial Intelligence and bots in coding/project building, it is crucial to identify which activity on the platform is organic and what is generated by bots. We aim to answer the following questions through our project:

- who are the bots?
- what repos do they target?
- can unlabeled humans behave suspiciously too?
- do multiple accounts move in coordinated patterns?
- what repo-level risk signals appear alongside this activity?

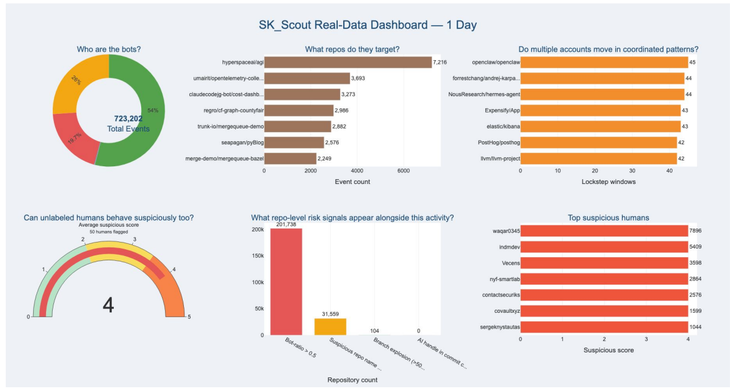


Figure 1 Report link: https://skscout-d1.netlify.app/



Figure 2 Report link: https://skscout-d3.netlify.app/

WORKFLOW

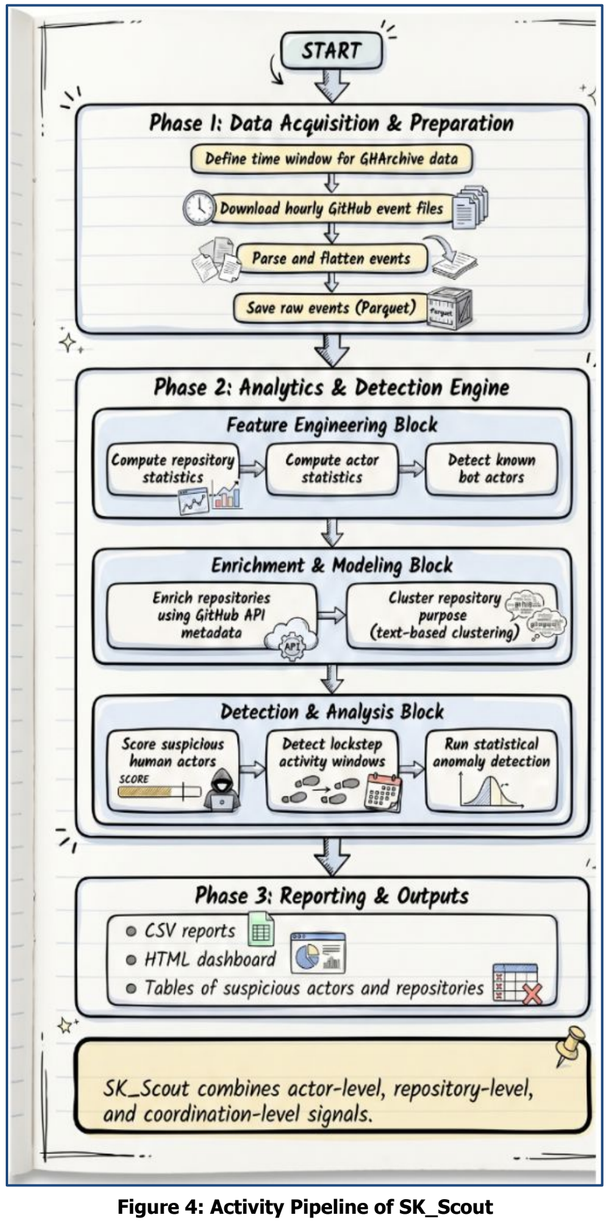


Figure 4: Activity Pipeline of SK_Scout

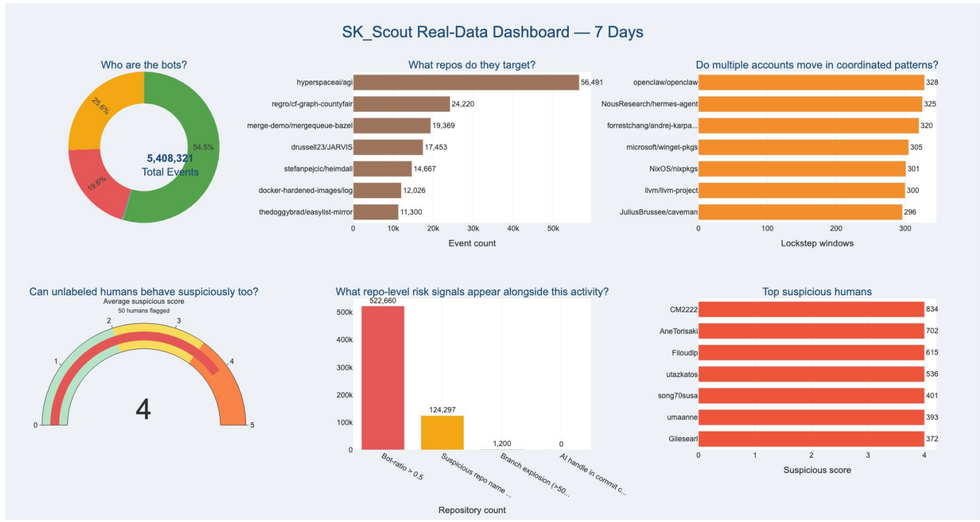


Figure 3 Report link: https://skscout-d7.netlify.app/

HOW IT STARTED AND WHERE IT ENDED

- Inspired by the ***Star_Scout***[1] that explored the popularity manipulation on GitHub, we wanted to look for deeper patterns interms of actors, repositories and time intervals.
- This led to a project that identifies bot-heavy behavour, suspicious human activity, coordinated lock-step interactions and high-risk repositories using behavioural, temporal and statistical signals.
- **Hypothesis**: Bot-heavy and suspicious repositories exhibit identifiable repository-level signatures, including high bot ratios, abnormal event concentration, unusual branching behavior, phishing-like naming patterns, and anomalous contribution traces.
- We conducted short term test for data gathered over 1 day, 3 days and 1 week to analyze the different patterns and their behaviours.
- ***Results*** indicate that GitHub's activity scales exponentially with bot and suspicious humans taking up a significant part of the event profiles
- In future, we wish to expand our scope to identify bots/suspicious human behaviour on **LinkedIn**'s job branch AND/OR identifying suspicious activities on **X (Twitter)** used for popularity and engagement.

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D1 D3 D7 GitHub

[1]H. He, H. Yang, P. Burckhardt, A. Kapravelos, B.Vasilescu, and C.Kästner. 2026. Six Million (Suspected) Fake Stars on GitHub: A Growing Spiral of Popularity Contests, Spam, and Malware. In 2026 IEEE/ACM 48th International Conference on Software Engineering (ICSE '26), April 12–18, 2026, Rio de Janeiro, Brazil. ACM, New York, NY, USA, 16 pages. https://doi.org/10.1145/3744916.3764531 https://github.com/hehao98/starscout